YAKEEN

NEET 2.0 2027

Physics | Lecture - 5

Units and Measurement

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Todays Goal

⇒ JEE Advance Question.

⇒ Significat digit

Accuracy & Precission.

(Q) Kitne logo ne H/W Try kiya ??

(a) yes

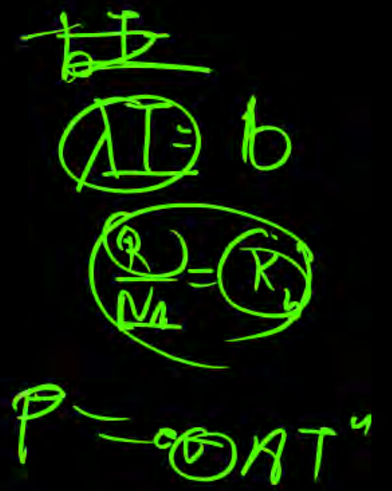
(b) NO

37. Consider that σ_s, k_B, b represent Stefan-Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-1} b$ is:

(1) $[L^{-1}T^{-1}K^{-2}]$ (2) $[L^{-1}K^{-2}]$

(3) $[L^{-1}T^{-1}K^{-3}]$ (4) $[L^{-1}T^{-1}K^{-4}]$

RE-NEET 2026



Question

A balloon is made of a material of surface tension S and its inflation outlet (from where gas is filled in it) has small area A. It is filled with a gas of density ρ and takes a spherical shape of radius R. When the gas is allowed to flow freely out of it, its radius r changes from R to 0 (zero) in time T. If the speed of gas coming out of the balloon depends on r as r^a and $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$ then

NEET-2025

1. $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -1, \gamma = +1, \delta = \frac{3}{2}$
2. $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = -\frac{1}{2}, \delta = \frac{5}{2}$
3. $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$
4. $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -\frac{1}{2}, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

$S = \frac{F}{A} = MT^{-2}$

$V = r^a$

$T = S^\alpha A^\beta \rho^\gamma$

$M^0L^0T^1 = (MT^{-2})^\alpha (L^2)^\beta (ML^{-3})^\gamma$

$\frac{1}{\sqrt{m}} \cdot \frac{\sqrt{m}}{\sqrt{m}} = \frac{1}{m}$

α, β, γ ka Power

AIPMT-2014 (Same)

Question

The relation between $[E]$ and $[B]$ is

[JEE Adv, 2018]

1. $[E] = [B][L][T]$
2. $[E] = [B][L]^{-1}[T]$
3. $[E] = [B][L][T]^{-1}$
4. $[E] = [B][L]^{-1}[T]^{-1}$

Question

The relation between $[\varepsilon_0]$ and $[\mu_0]$ is

[JEE Adv, 2018]

1. $[\mu_0] = [\varepsilon_0][L]^2[T]^{-2}$
2. $[\mu_0] = [\varepsilon_0][L]^{-2}[T]^2$
3. $[\mu_0] = [\varepsilon_0]^{-1}[L]^2[T]^{-2}$
4. $[\mu_0] = [\varepsilon_0]^{-1}[L]^{-2}[T]^2$

Question

A physical quantity $\vec{S}$ is defined as $\vec{S} = (\vec{E} \times \vec{B})/\mu_0$, where $\vec{E}$ is electric field, $\vec{B}$ is magnetic field and μ_0 is the permeability of free space. The dimensions of $\vec{S}$ are the same as the dimensions of which of the following quantity (ies)? **(JEE Adv. 2021)**

1. $\dfrac{\text{Energy}}{\text{Charge} \times \text{Current}}$
2. $\dfrac{\text{Force}}{\text{Length} \times \text{Time}}$
3. $\dfrac{\text{Energy}}{\text{Volume}}$
4. $\dfrac{\text{Power}}{\text{Area}}$

$S = \dfrac{E \times B}{\mu}$

$= \dfrac{F \times F}{q \quad Il} \times \dfrac{I^2 I}{F}$

$= \dfrac{FI}{ql} = \dfrac{F\cancel{I}}{\cancel{I}tl} = \dfrac{F}{tl}$

$= \dfrac{F \times l}{tll} = \dfrac{\text{energy}}{l^2 t} = \dfrac{\text{Power}}{l^2(\text{Area})}$

$F = \mu_0 I^2$

$F = qE \qquad F = BIl$

$E = \dfrac{F}{q} \quad -(i)$

$B = \dfrac{F}{Il} \quad -(ii)$

$\mu_0 = \dfrac{F}{I^2}$

Question

2010 → 2021 AIEEE

Let us consider a system of units in which mass and angular momentum are dimensionless. If length has dimension of L, which of the following statement(s) is/are correct? **[JEE Adv, 2019]**

Mass $= M^0L^0T^0 = 1$

$L = p \times v = m^2 L^2 T^{-1} = M^0L^0T^0 = 1$

$1\ L^2T^{-1} = 1$

$L^2 = T$ — (1) **

1. The dimension of force is L^{-3}

$F = ma \quad LT^{-2} = L(L^2)^{-2} = L \times L^{-4} = L^{-3}$

2. The dimension of energy of L^{-2}

$E = ma \quad L^2T^{-2} = L^2(L^2)^{-2} = L^2 \times L^{-4} = L^{-2}$

3. The dimension of power is L^{-5}

$P = ma \quad L^2T^{-3} = L^2L^{-6} = L^{-4}$

4. The dimension of linear momentum is L^{-1}

$P = mv \quad L^2T^{-1}$

$L \times L^{-2} = L^{-1}$

$P \ (P = L^{-4})$

Question

Expression for time in terms of G (universal gravitational constant), h (Planck constant) and c (speed of light) is proportional to

Let n

1. $\sqrt{\frac{hc^3}{G}} = \sqrt{\frac{M}{M^{-1}}} = \sqrt{M^2}$ ✗

2. $\sqrt{\frac{c^3}{Gh}} = \sqrt{\frac{L^3}{M^{-1}M L^3 \times L^2}}$ ✗

3. $\sqrt{\frac{Gh}{c^5}} = \sqrt{\frac{L^3 \times L^2}{L^5}} = \frac{L^5}{L^5} = L^0$ ✓

4. $\sqrt{\frac{Gh}{c^3}} = \sqrt{\frac{L^3 \times L^2}{L^3}} = L$ ✗

$$M^a L^0 t = h^x G^y c^z$$

Question

In a new system of units energy (E), density (d) and power (P) are taken as fundamental units, then the dimensional formula of universal gravitational constant G will be

done JEE Adv – 2012

1. $[E^{-1} d^{-2} P^{2}]$
2. $[E^{-2} d^{-1} P^{2}]$
3. $[E^{2} d^{-1} P^{-1}]$
4. $[E^{1} d^{-2} P^{-2}]$

Question

In terms of potential difference V, electric current I, permittivity ε_0, permeability μ_0 and speed of light c the dimensionally correct equations is/are **[JEE Adv, 2015]**

1. $\mu_0 I^2 = \varepsilon_0 V^2$
2. $\varepsilon_0 I = \mu_d V$
3. $I = E_0 c V$
4. $\mu_0 c I = \varepsilon_0 V$

hint: $\left(R = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \frac{V}{I}\right)$ use $(V = IR)$

$\frac{\mu_0}{\varepsilon_0} = \frac{V^2}{I^2}$

$\mu_0 I^2 = \varepsilon_0 V^2$

Question

Try at home

hint → Sirf length Ka Power Socho

Young's modulus of elasticity Y is expressed in terms of three derived quantities, namely, the gravitational constant G. Planck's constant h and the speed of light c, as $Y = c^{\alpha} h^{\beta} G^{\gamma}$. Which of the following is the correct option? **[JEE Adv, 2023]**

1. $\alpha = 7, \beta = -1, \gamma = -2$
2. $\alpha = -7, \beta = -1, \gamma = -2$
3. $\alpha = 7, \beta = -1, \gamma = 2$
4. $\alpha = -7, \beta = 1, \gamma = -2$

$G^{\gamma} = M^{-1} L^{3} T^{-2}$

$h^{\beta} = M L^{2} T^{-1}$

$C^{\alpha} = L T^{-1}$

$7 - 2 - 6$

$7 - 8 = -1$

$(Y = M L^{-1} T^{2})$

$Y = G^{\gamma} h^{\beta} C^{\alpha}$

$L^{-1} = (L^{3})^{\gamma} (L^{2})^{\beta} (L)^{\alpha}$

$3\alpha + 2\beta + \alpha = -1$

$3(-2) + 2(-2) + 7 = -1$

Question

TYY

A length-scale (l) depends on the permittivity (ε) of a dielectric material, Boltzmann constant (k_B) ,the absolute temperature (T), the number per unit volume (n) of certain charged particles, and the charge (q) carried by each of the particles. Which of the following expression(s) for l is (are) dimensionally correct?

[JEE Advanced 2016]

1. $l = \sqrt{\left(\frac{nq^2}{\varepsilon k_B T}\right)}$
2. $l = \sqrt{\left(\frac{\varepsilon k_B T}{nq^2}\right)}$
3. $l = \sqrt{\left(\frac{q^2}{\varepsilon n^{2/3} k_B T}\right)}$
4. $l = \sqrt{\left(\frac{q^2}{\varepsilon n^{1/3} k_B T}\right)}$

Question

Sometimes it is convenient to construct a system of units so that all quantities can be expressed in terms of only one physical quantity. In one such system, dimensions of different quantities are given in terms of a quantity x as follows: [position] = $[x^{\alpha}]$; speed = $[x^{\beta}]$; [acceleration] = $[x^{p}]$; [linear momentum] = $[x^{q}]$; [force = $[x^{r}]$. Then

[JEE Adv, 2020]

1. $\alpha + p = 2\beta$
2. $p + q - r = \beta$
3. $p - q + r = \alpha$
4. $p + q - r = \beta$

Position $= x^{\alpha}$

speed $= x^{\beta} = \frac{\text{Position}}{\text{time}}$

$x^{\beta} = \frac{x^{\alpha}}{t}$

$t = \frac{x^{\alpha}}{x^{\beta}} = x^{\alpha-\beta}$

$acc = \frac{v}{t} = x^{p}$

$\frac{x^{\beta}}{x^{\alpha-\beta}} = x^{p}$

$x^{\beta-\alpha+\beta} = x^{p}$

$2\beta - \alpha = p$

$2\beta = \alpha + \beta$

$p = mv = x^{q}$ — (II)

$F = x^{r}$ — (III)

Question

Kis liye ye question kiya → clarity, calculate, learn, think

Sometimes it is convenient to construct a system of units so that all quantities can be expressed in terms of only one physical quantity. In one such system, dimensions of different quantities are given in terms of a quantity x as follows: [position] = $[x^{\alpha}]$; speed = $[x^{\beta}]$; [acceleration] = $[x^{p}]$; [linear momentum] = $[x^{q}]$; [force = $[x^{r}]$. Then

[JEE Adv, 2020]

1. $\alpha + p = 2\beta$ (position, acc, speed)
2. $p + q - r = \beta$
3. $p - q + r = \alpha$
4. $p + q - r = \beta$

Position $= x^{\alpha}$

speed (velocity) $= x^{\beta} = \frac{\text{Position}}{\text{time}}$

$x^{\beta} = \frac{x^{\alpha}}{t}$

$t = \frac{x^{\alpha}}{x^{\beta}} = x^{\alpha-\beta}$

$acc = \frac{v}{t} = x^{p}$

$t = \frac{v}{x^{p}} = \frac{x^{\beta}}{x^{p}}$

$p = mv = x^{q}$ — (II)

$F = x^{r}$ — (III)

$F = \frac{p}{t}$

$x^{r} = \frac{x^{q}}{x^{\beta-p}}$

$x^{r} = x^{2-\beta+p}$

$r = 2 - \beta + p$

Question

Try at home

In a particular system of units, a physical quantity can be expressed in terms of the electric charge e, electron mass m_e, Planck's constant h, and coulomb's constant $k = \frac{1}{4\pi\varepsilon_0}$, where ε_0 is the permittivity of vacuum. In terms of these physical constants, the dimension of the magnetic field is $[B] = [e]^{\alpha}[m_e]^{\beta}[h]^{\gamma}[k]^{\delta}$. The value of $\alpha + \beta + \gamma + \delta$ is

(JEE Adv. 2022)

Question

The frequency of vibrations f of a mass m suspended from a spring of spring constant K is given by a relation of type $f = cm^x K^y$, where c is a dimensionless constant. The values of x and y are:

1. $x = \frac{1}{2}, y = \frac{1}{2}$
2. $x = \frac{-1}{2}, y = \frac{-1}{2}$
3. $x = \frac{1}{2}, y = \frac{-1}{2}$
4. $x = \frac{-1}{2}, y = \frac{1}{2}$

$f = m^x K^y$

$M^0 L^0 T^{-1} = m^x \left(\frac{F}{L}\right)^y$

$M^0 L^0 T^{-2} = m^x (MT^{-2})^y$

$(MT^{-2})^{-\frac{1}{2}}$

$+1$

Question

If speed V, area A and force F are chosen as fundamental units, then the dimension of Young's modulus will be: **[JEE Main 2020]**

1. $FA^{-1} V^{0}$
2. $FA^{2} V^{-1}$
3. $FA^{2} V^{-2}$
4. $FA^{2} V^{-3}$

Question

If momentum (P), area (A) and time (T) are taken to be the fundamental quantities then the dimensional formula for energy is **[JEE Main 2020]**

1. $[P^{1/2} AT^{-1}]$
2. $[P^2 AT^{-2}]$
3. $[PA^{1/2} T^{-1}]$
4. $[P^{-1} AT^{-2}]$

$E = P^n A T$

$\rightarrow M L^2 T^{-2}$

Question

If force (F), velocity (V) and time (T) are considered as fundamental physical quantity, then dimensional formula of density will be: **[JEE Main 2023]**

density =

1. FV^4T^{-6}
2. $FV^{-4}T^{-2}$
3. $F^2V^{-2}V^6$
4. $FV^{-2}T^2$

Question

If momentum [P], area [A] and time [T] are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is:

1. $[P^{1} A^{-1} T^{0}]$
2. $[P^{1} A^{1} T^{-1}]$
3. $[P^{1} A^{-1} T]$
4. $[P^{1} A^{-1} T^{-1}]$

$F = 6\pi\eta r v_T$

$\eta = \frac{F}{r \times v_T} = M L^{-1} T^{-1} = \frac{M^{1}/T^{1}}{L^{2}} = M L^{-1} T^{1}$

Question

Time, velocity and angular momentum then find dimension of mass.

[JEE Main 2021]

Question

Planck's constant (h), Current (I) speed of light (c) correct dimension of stopping potential

[JEE Main 2020]

1. $hI^{-1}G^{-1}c^{5}$
2. $h^{-1}I^{1}G^{-1}c^{6}$
3. $h^{0}I^{-1}G^{-1}c^{6}$
4. $h^{0}I^{1}G^{-1}c^{6}$

Question

Sometimes it is convenient to construct a system of units so that all quantities can be expressed in terms of only one physical quantity. In one such system, dimensions of different quantities are given in terms of a quantity x as follows: [position] = $[x^{\alpha}]$; speed = $[x^{\beta}]$; [acceleration] = $[x^{p}]$; [linear momentum] = $[x^{q}]$; [force = $[x^{r}]$. Then **[JEE Adv, 2020]**

1. $\alpha + p = 2\beta$
2. $p + q - r = \beta$
3. $p - q + r = \alpha$
4. $p + q - r = \beta$

copy at home

LIMITATION OF DIMENSIONAL ANALYSIS

$T = \sqrt{\frac{l}{g}}$ ✓ $T = 2\pi\sqrt{\frac{l}{g}}$ ✗

(1) It is not use to derive dimensionless proportional constant.

(2) It can not derive dimensionless function, like $\sin\theta$, $\cos\theta$, $\tan\theta$, e^x etc. (Angle)

(3) If physical quantity depends upon two P.Q. of same dimension

(4) It can not derive formula which have 't' and '–' term

Ex : $S = ut + \frac{1}{2}at^2$

(5) We equate the power of M, L and T. So, it only work when quantity depends only on three physical quantity. in mechanics.

Ex : If force depends upon energy, velocity, time work.

(1) If Torque, energy and work consider as fundament P.Q then find dimension of time.

Soln Not Possible

(2) If acc^n due to gravity depend on force, energy, mass & velocity. then find dimension of acc^n due to gravity.

$$\text{Acc}^n = F^{\alpha} E^{\beta} M^{\gamma} V^{z}$$

$$M^0 L^1 T^{-2} = (MLT^{-2})^{\alpha} (ML^2T^{-2})^{\beta} M^{\gamma} (LT^{-1})^{z}$$

(3) If force depends on Angle, velocity & energy then find dimⁿ of force in terms of them.

$$\text{Force} = \theta^{\alpha} V^{\beta} E^{\gamma}$$

Not Possible

Question

Which of the following equation can be derived dimensionally

1. $s = vt - \frac{1}{2}at^2$
2. $v^2 = u^2 - 2as$
3. $h = \frac{\omega^2 r^2}{2g}$
4. $v = \frac{d}{t}$

Question

Which of the following equation can not be derived dimensionally

1. $F = 6\pi r\eta$ → Ans (1) (Best Possible)
   → dimension.
2. $\theta = \omega t$
3. $\frac{d\theta}{dt} = \rho A v$
4. $P = \rho g h$

$\theta = \omega t$

$\theta \propto \omega^{\alpha} t^{\beta}$

$M^0L^0T^0 = (T^{-1})^{\alpha}(T)^{\beta}$

$M^0L^0T^0 = T^{-\alpha+\beta}$ $\quad -\alpha+\beta=0$

$\alpha=\beta$





all customer shifted from M.R Sir to Kallua shop

$M_1 = 1\ kg$

$M_2 = 2\ kg$

$M_3 = 3\ kg$

$M_1 = 1.00\ kg$

$M_2 = 2.00\ kg$

- **Significant figure (digit) = (All certain + one uncertain)**

Ex. : **M = 256703 kg** ⇒ 6 significant digit ⇒ 5 certain + 1 uncertain digit

Significant digit indicate precision in the measurement.

- **Choice of change in different unit does not affect the significant digit.**

$2.5\,kg \longrightarrow 2.5 \times 10^3\ gm$

copy at ghar

Rule to find significant digit :

1. All non-zero are significant. ⟶ 2457m ⟶ 4 (significant)
2. all zero between non-zero are significant. ⟶ 24507 ⟶ 5
3. Trailing zero without decimal point is not significant. 4500cm ⟶ 2
4. After decimal place all zero are significant 45.00cm ⟶ 4
5. If number is less than one (1), all zero left of non-zero are not significant.
   0.02m ⟶ 1
   0.0307m ⟶ 3
6. Exact number have infinite significant number.
   Ex ⟶ 5 Pen ⟶ infinite
7. In power form power is not significant.
   2.303×10^{23} m ⟶ 4

Counting of Significant Digit

Counting of Significant Digit

(1) 23.500 m

(2) 402 cm

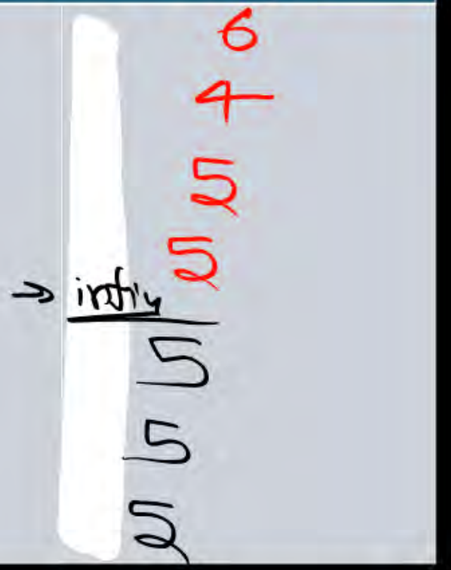
(3) 5280 kg

(4) 6.780

(5) 0.034 gram

(6) 3.070 gram

(7) 1.608×10^{23}



Question

Give the number of significant figures in each measurement.

1. 36.7 m → 3
2. 0.006606 s → 4
3. 2,002 kg → 4
4. 306,490,000 people infinite

Question

Given P = 0.0030 m, Q = 2.40 m and R = 3000 m, the number of significant figures in P, Q, R are respectively:

1. 1, 2, 1
2. 2, 3, 1
3. 4, 2, 1
4. 4, 2, 4

Question

Give the number of significant figures in each measurement.

1. 36.7 m
2. 0.006606 s
3. 2,002 kg
4. 306,490,000 people

- **Addition or Subtraction** with significant digit
  - Final result is written in minimum decimal places.

$$\begin{array}{r} 4.71 \\ +\ 1.3 \\ \hline 6.01 \end{array} \Rightarrow 6.0$$

$$\begin{array}{r} 3.813 \\ +\ 2.21 \\ \hline = 6.023 \end{array} \Rightarrow = 6.02$$

- **Multiplication or Division**
  - Final result written in minimum significant figure.

$$3.01 \times 3.1$$

Question

Copy

The area of rectangle of length 55.3 m and breadth 25 m. [NEET-2022]

1. 1382
2. 1382.5
3. 14×10^2
4. 138×10^1

$A = l \times b$

$= 55.3 \times 25$

Question

Taking into account of significant digit. What is the value of

9.99 m – 0.0099 m.

[NEET-2020]

1. 9.88 m
2. 9.890 m
3. 9.9 m
4. 9.9801 m

Question

Add these three length:

$l_1 = 0.307$ m, 0.52 m and 0.4 m

1. 1.22 m
2. 1.2 m
3. 1.3 m
4. 1.7 m

ROUNDING OFF

- If digit to be removed is less than 5 then there is no change in parimary number.
- If digit to be removed is greater than 5 then there previous number increases by 1.
- If digit is 5, then previous number remains same if even and increase by 1 then if odd.

Question

Express the final answer to the proper number of significant figures.

(i) $101.2 + 18.702 = ?$

(ii) $202.88 - 1.013 = ?$

Question

Calculate the correct number of significant figures for the final solution:

(i) Round 4.7475 to 4 significant figures

(ii) Round 4.7465 to 4 significant figures

Question

If A = 2.413 and B = 1.2 then find A + B.

Question

If A = 25.5 and B = 5 then find $\frac{A}{B}$.

Question

If true value of length is 6.57 m then which of the following reading is most accurate and most precise.

1. 6.52 m
2. 6.61 m
3. 6.513 m
4. 6.68 m

Question

Which of the following reading is most accurate? **[NEET 2024]**

1. 2.4 m
2. 2.41 m
3. 2 m
4. 2.413 m

Question

If absolute error and actual value of a number are 5, 15 respectively then relative error is

Question

If $l_1 = (10 \pm 2)$ cm and $l_2 = (20 \pm 1)$ cm. Find $l = l_1 + l_2$ with error.

Question

In a series of successive measurements in an experiment, the readings of the period of oscillation of a simple pendulum were round to be 2.63s, 2.56s, 2.42s, 2.71s and 2.80s. Calculate

(i) The mean value of the period of oscillation

(ii) The absolute error in each measurement

(iii) The mean absolute error

(iv) The relative error

(v) The percentage error.

Thank

You